

Vector Autoregression Estimates

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Date: 04/28/26 Time: 17:47

Sample (adjusted): 5/10/2016 3/30/2026

Included observations: 2453 after adjustments

Standard errors in () & t-statistics in []

	BRENT_RET	BRENT_VOL	WTI_RETURN	WTI_VOL	BRENT_VOL	WTI_VOLUM
BRENT_RETURN(-1)	0.025961 (0.05913) [0.43904]	0.038190 (0.06886) [0.55464]	0.259415 (0.06458) [4.01714]	0.554712 (0.07525) [7.37204]	-0.712872 (1.28885) [-0.55311]	-1.001679 (1.03035) [-0.97218]
BRENT_RETURN(-2)	-0.014579 (0.06193) [-0.23541]	-0.086426 (0.07212) [-1.19839]	0.099839 (0.06764) [1.47610]	0.132804 (0.07881) [1.68510]	-2.776681 (1.34992) [-2.05692]	1.164839 (1.07917) [1.07939]
BRENT_RETURN(-3)	-0.081874 (0.06210) [-1.31848]	-0.179131 (0.07231) [-2.47725]	-0.073101 (0.06782) [-1.07792]	-0.410893 (0.07902) [-5.19983]	-1.673543 (1.35351) [-1.23645]	0.209650 (1.08204) [0.19375]
BRENT_RETURN(-4)	0.068553 (0.05887) [1.16453]	-0.212620 (0.06855) [-3.10174]	0.039663 (0.06429) [0.61695]	-0.190802 (0.07491) [-2.54709]	0.685156 (1.28310) [0.53398]	0.220093 (1.02575) [0.21457]
BRENT_VOL(-1)	0.017100 (0.02676) [0.63893]	1.001191 (0.03117) [32.1251]	0.010430 (0.02923) [0.35684]	0.238720 (0.03406) [7.00935]	-1.247741 (0.58336) [-2.13890]	-0.421096 (0.46635) [-0.90295]
BRENT_VOL(-2)	0.035244 (0.03528) [0.99887]	-0.089387 (0.04109) [-2.17559]	0.036152 (0.03853) [0.93820]	-0.318115 (0.04490) [-7.08513]	0.505623 (0.76906) [0.65746]	0.177938 (0.61481) [0.28942]
BRENT_VOL(-3)	-0.062458 (0.03507) [-1.78103]	0.001828 (0.04084) [0.04476]	-0.076922 (0.03830) [-2.00848]	-0.130037 (0.04463) [-2.91396]	0.476327 (0.76438) [0.62316]	-0.137772 (0.61107) [-0.22546]
BRENT_VOL(-4)	-0.000567 (0.02652) [-0.02139]	0.054668 (0.03088) [1.77037]	0.017362 (0.02896) [0.59951]	0.300766 (0.03374) [8.91296]	0.078505 (0.57800) [0.13582]	0.404017 (0.46207) [0.87436]
WTI_RETURN(-1)	-0.016211 (0.05411) [-0.29960]	-0.029807 (0.06301) [-0.47307]	-0.266996 (0.05909) [-4.51830]	-0.580801 (0.06885) [-8.43521]	0.668343 (1.17938) [0.56669]	1.163386 (0.94283) [1.23392]
WTI_RETURN(-2)	0.028615 (0.05767) [0.49623]	0.029888 (0.06715) [0.44510]	-0.088775 (0.06298) [-1.40965]	-0.205619 (0.07338) [-2.80208]	2.092493 (1.25691) [1.66479]	-1.521736 (1.00481) [-1.51444]
WTI_RETURN(-3)	0.029767 (0.05788) [0.51432]	0.125417 (0.06740) [1.86088]	0.009164 (0.06321) [0.14498]	0.318521 (0.07365) [4.32474]	1.968148 (1.26154) [1.56012]	-0.147552 (1.00851) [-0.14631]
WTI_RETURN(-4)	-0.051653 (0.05486) [-0.94161]	0.209818 (0.06388) [3.28464]	-0.019003 (0.05991) [-0.31720]	0.151911 (0.06981) [2.17618]	-0.607115 (1.19569) [-0.50775]	-0.210103 (0.95587) [-0.21980]
WTI_VOL(-1)	0.001758 (0.02383) [0.07379]	0.139729 (0.02775) [5.03610]	0.008514 (0.02602) [0.32719]	0.861497 (0.03032) [28.4133]	0.970201 (0.51934) [1.86813]	-0.350122 (0.41518) [-0.84330]
WTI_VOL(-2)	-0.054538 (0.02994) [-1.82166]	-0.043237 (0.03486) [-1.24022]	-0.073260 (0.03270) [-2.24062]	0.241518 (0.03810) [6.33943]	-0.902392 (0.65256) [-1.38285]	0.653629 (0.52168) [1.25294]

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WTI_VOL(-3)	0.016050 (0.02987) [0.53742]	0.002275 (0.03478) [0.06542]	0.060343 (0.03262) [1.85012]	0.128438 (0.03800) [3.37959]	-0.312652 (0.65096) [-0.48029]	-0.406565 (0.52040) [-0.78126]
WTI_VOL(-4)	0.046966 (0.02340) [2.00733]	-0.085589 (0.02725) [-3.14143]	0.008863 (0.02555) [0.34685]	-0.323368 (0.02977) [-10.8610]	0.328002 (0.50998) [0.64317]	-0.013057 (0.40769) [-0.03203]
BRENT_VOLUME_CHA	-0.001034 (0.00094) [-1.10488]	0.003036 (0.00109) [2.78652]	0.000498 (0.00102) [0.48760]	0.001400 (0.00119) [1.17588]	-0.450279 (0.02039) [-22.0801]	-0.026183 (0.01630) [-1.60604]
BRENT_VOLUME_CHA	0.002173 (0.00101) [2.16184]	-0.001526 (0.00117) [-1.30361]	0.002395 (0.00110) [2.18192]	-0.000108 (0.00128) [-0.08441]	-0.289567 (0.02191) [-13.2162]	0.001165 (0.01752) [0.06651]
BRENT_VOLUME_CHA	0.000556 (0.00101) [0.55281]	-8.78E-05 (0.00117) [-0.07491]	0.000442 (0.00110) [0.40254]	0.000903 (0.00128) [0.70528]	-0.229607 (0.02194) [-10.4665]	-0.005659 (0.01754) [-0.32270]
BRENT_VOLUME_CHA	-8.15E-05 (0.00094) [-0.08657]	-0.000539 (0.00110) [-0.49223]	-0.000365 (0.00103) [-0.35541]	-0.000395 (0.00120) [-0.33009]	-0.131631 (0.02051) [-6.41801]	0.008478 (0.01640) [0.51709]
WTI_VOLUME_CHANG	-0.001062 (0.00117) [-0.90470]	-0.000783 (0.00137) [-0.57269]	-0.000971 (0.00128) [-0.75716]	-0.001096 (0.00149) [-0.73396]	-0.013882 (0.02559) [-0.54258]	-0.201495 (0.02045) [-9.85139]
WTI_VOLUME_CHANG	-0.000166 (0.00118) [-0.14128]	0.001474 (0.00137) [1.07594]	-0.000387 (0.00129) [-0.30078]	-1.66E-05 (0.00150) [-0.01106]	-0.016866 (0.02565) [-0.65750]	-0.380136 (0.02051) [-18.5371]
WTI_VOLUME_CHANG	-2.91E-05 (0.00118) [-0.02479]	-0.000332 (0.00137) [-0.24275]	0.000179 (0.00128) [0.13933]	-0.001281 (0.00150) [-0.85673]	-0.072917 (0.02562) [-2.84629]	-0.195463 (0.02048) [-9.54403]
WTI_VOLUME_CHANG	0.000475 (0.00117) [0.40564]	0.001565 (0.00136) [1.14864]	0.000140 (0.00128) [0.10943]	0.001228 (0.00149) [0.82495]	-0.066385 (0.02550) [-2.60340]	-0.146568 (0.02039) [-7.18994]
C	0.000223 (0.00120) [0.18645]	0.005436 (0.00139) [3.90251]	0.002522 (0.00131) [1.93060]	0.001751 (0.00152) [1.15016]	0.031816 (0.02607) [1.22032]	0.029667 (0.02084) [1.42341]
D_COVID	-0.000154 (0.00159) [-0.09724]	0.001319 (0.00185) [0.71326]	0.000282 (0.00173) [0.16280]	0.004177 (0.00202) [2.06703]	0.010877 (0.03461) [0.31427]	0.007715 (0.02767) [0.27885]
D_IRAN	0.000168 (0.00118) [0.14253]	0.000507 (0.00137) [0.36965]	0.000191 (0.00129) [0.14855]	0.001386 (0.00150) [0.92503]	-0.010376 (0.02567) [-0.40419]	-0.005315 (0.02052) [-0.25899]
D_RUSSIA	-0.000325 (0.00140) [-0.23168]	0.001226 (0.00163) [0.75022]	0.000689 (0.00153) [0.44977]	0.000636 (0.00179) [0.35637]	0.007676 (0.03058) [0.25100]	0.009228 (0.02445) [0.37742]
R-squared	0.014111	0.972918	0.020177	0.977880	0.193232	0.155204
Adj. R-squared	0.003134	0.972616	0.009268	0.977634	0.184250	0.145798
Sum sq. resids	1.329350	1.802577	1.585512	2.152651	631.5666	403.6272
S.E. equation	0.023413	0.027264	0.025570	0.029794	0.510333	0.407976
F-statistic	1.285512	3226.584	1.849553	3970.614	21.51189	16.50058
Log likelihood	5743.086	5369.584	5526.956	5151.901	-1816.463	-1267.345
Akaike AIC	-4.659670	-4.355144	-4.483454	-4.177661	1.503843	1.056131
Schwarz SC	-4.593408	-4.288881	-4.417191	-4.111398	1.570105	1.122393

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Mean dependent	0.000301	0.326102	0.000114	0.351844	-0.000620	-0.001319
S.D. dependent	0.023450	0.164758	0.025689	0.199223	0.565035	0.441422
Determinant resid covariance (dof adj.)	4.68E-16					
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Log likelihood	22494.24					
Akaike information criterion	-18.20322					
Schwarz criterion	-17.80564					
Number of coefficients	168					